DEPARTMENT OF DEFENSE
HUMAN RESOURCES ACTIVITY
DEFENSE MANPOWER DATA CENTER
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ALEXANDRIA, VA 22350-4000

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MEMORANDUM FOR SECRETARIES OF THE MILITARY DEPARTMENTS
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SUBJECT: Changes to the Two-Dimensional Barcode on DoD Identification Cards

Pursuant to DoD Instruction 1000.30 (Reduction of Social Security Number (SSN) Use Within DoD) and redesign of the DoD identification cards issued to family members, retirees, and beneficiaries, DoD will be changing the two-dimensional barcode (i.e., Portable Data File (PDF417) on all newly issued DoD identification cards beginning in October 2017. These changes will be reflected as old cards expire and include the following:

- Removal of the SSN from PDF417 of all DoD identification cards
- Reformatting of the PDF417 for all DoD identification cards
- Removal of the facial image from PDF417 of all DoD identification cards for family members, retirees, and beneficiaries.

The attached bulletin provides the technical detail of these changes. Any organizations that may be using the PDF417 barcode on DoD identification cards are strongly encouraged to ensure their systems are updated to accommodate these changes. My point of contact for this effort is Mr. Dan T. Ma (Credentialing Division, dan.t.ma.civ@mail.mil, (831) 583-2400 ext. 4524).

Mary Snavely-Dixon

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Director

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RE: Technical Bulletin—Implementation of SSN Removal from DoD ID Card Barcodes

1. Introduction

As part of the Department's Social Security Number (SSN) reduction activity (i.e., Under Secretary of Defense for Personnel and Readiness Memorandum, "Updated Plan for the Removal of Social Security Numbers (SSNs) from Department of Defense (DoD) Identification (ID) Cards," 5 November 2010, DoD will remove the SSN from barcodes that are printed on the Common Access Card (CAC) and the remaining family of DoD identification cards. DoD completed each of the initial phases of the transition; first, by removing the SSN on the printed surface and second, by removing the SSN embedded in the Code 39 barcode. This next phase will remove the SSN from the Portable Data File (PDF417) barcode. The purpose of this document is to describe how the Department will implement removing the SSN from the PDF417 barcode.

2. Audience & Scope

DMDC has written this document for the community of users, application developers and system maintainers who interact with the barcodes on DoD ID cards. This document will outline the future format of the PDF417 barcode on all DoD ID cards.

3. Description of the Change

DMDC removed the SSN field from the structure of the PDF417 barcode and realigned the remaining fields. Applications using the SSN within the barcode should already be aware of the migration to the Electronic Data Interchange Person Identifier (EDIPI) as the unique identifier for individuals within the DoD. System maintainers are expected to modify their systems to use the EDIPI instead of the SSN. This section will illustrate further changes to the CAC, Sponsor ID, and Dependent ID.

CAC PDF417 CHANGES

Additions to the CAC PDF417:

(U/FOUO)

Additions to the CAC Barcode	**Size**	**Description**
DoD Benefits Number	8	8 digit base 32 number ranging from 00000000000 – 99999999999
Country of Citizenship	3	A 2 character ISO country code, which will be updated to 3 characters in the future
Unique Card Identifier	6	6 digit base 32 number ranging from 999000000 – 999999999

Eliminations to the CAC PDF417:

(U/FOUO)

Eliminations to the CAC Barcode	**Size**	**Description**
Person Designator Identifier (PDI)	6	9 digit PDI in the range of 000-00-0000 to 999-99-9999 was stored as a 6 digit base 32 number
Person Designator Type Code	1	The code that represents the type of Person Designator. Ex. Social Security Number

SPONSOR ID PDF417 CHANGES

Additions to the Sponsor ID PDF417:

(U/FOUO)

Additions to the Sponsor ID Barcode	**Size**	**Description**
Bar Code Version Code	1	"4"
DoD EDI Person Identifier (EDIPI)	7	7 digit base 32 number ranging from 1000000000 - 9999999999
DoD Benefits Number	8	8 digit base 32 number ranging from 00000000000 – 99999999999
Country of Citizenship	3	A 2 character ISO country code, which will be updated to 3 characters in the future
Personnel Category Code	1	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Personnel Entitlement Condition Type	2	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Pay Plan Code	2	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Unique Card Identifier	6	6 digit base 32 number ranging from 999000000 – 999999999
First Name	20	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname
Middle Initial	1	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname
Surname	26	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname

Eliminations to the Sponsor ID PDF417:

(U/FOUO)

Eliminations to the Sponsor ID Barcode	**Size**	**Description**
Digital Photograph		The compressed, resized JFIF photograph is no longer stored in the barcode

Identification Code	4	*HEADER DATA:* Indicates population category. See following tables
Ver	1	*HEADER DATA:* The version of the PDF417 bar code for the form type shown by the Identification Code
PDF417 Size	2	*HEADER DATA:* Length of PDF417 bar code following this field. This value plus 7 should equal the total length of the bar code read
Checksum	1	*HEADER DATA:* Checksum of the bytes in the bar code
RSize	1	*HEADER DATA:* Indicates the size of the RAPIDS text data contained in thePDF417 bar code
Person Designator Identifier (PDI)	6	9 digit PDI in the range of 000-00-0000 to 999-99-9999 was stored as a 6 digit base 32 number
Family Sequence Number	1	See Appendix D of DoD ID Card Bar Codes & Software Development Kit
Reserved for Future Use	9	9 character padding
DEERS Dependent Suffix/ DEERS Dependent Suffix Code	2	e.g. DDS: “20” / DDSC: “00”
Height	2	See DODI 1000.13
Weight	3	See DODI 1000.13
Hair Color	2	See DODI 1000.13
Eye Color	2	See DODI 1000.13
Direct Care Flag	1	S/N/R – See Appendix C of DoD ID Card Bar Codes & Software Development Kit
CHAMPUS Flag	1	M/N/F - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Commissary Flag	1	Y/N/X - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
MWR Flag	1	Y/N/X - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Exchange Flag	1	U/N/L/X - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
CHAMPUS Effective Date	9	YYYYMMDD Format
CHAMPUS Expiration Date	9	YYYYMMDD Format
Form Number	6	See Table 5 of DoD ID Card Bar Codes & Software Development Kit
Security Code	4	Machine generated
Service/component Code	1	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Status	6	See DODI 1000.13
Geneva Conventions Code	3	See DODI 1000.13
Blood Type	3	See DODI 1000.13

DEPENDENT ID PDF417 CHANGES

Additions to the Dependent ID PDF417:

(U/FOUO)

Additions to the Dependent ID Barcode	**Size**	**Description**
Bar Code Version Code	1	"4"
DoD EDI Person Identifier (EDIPI)	7	7 digit base 32 number ranging from 1000000000 - 9999999999
DoD Benefits Number	8	8 digit base 32 number ranging from 00000000000 – 99999999999
Country of Citizenship	3	A 2 character ISO country code, which will be updated to 3 characters in the future
Sponsor's DoD EDI Person Identifier (EDIPI)	7	7 digit base 32 number ranging from 1000000000 - 9999999999
Sponsor's Deceased Flag	1	Y or N to indicate if sponsor is deceased or not, originally obtained from Status field
Sponsor's Personnel Category	1	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Sponsor's Personnel Entitlement Condition Type	2	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Sponsor's Pay Plan Grade	2	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Unique Card Identifier	6	6 digit base 32 number ranging from 999000000 – 999999999
First Name	20	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname
Middle Initial	1	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname
Surname	26	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname
Sponsor's First Name	20	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname
Sponsor's Middle Initial	1	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname
Sponsor's Surname	26	The name field is now separated between 3 fields: First Name, Middle Initial, and Surname

Eliminations to the Dependent ID PDF417:

(U/FOUO)

Eliminations to the Dependent ID Barcode	**Size**	**Description**
Digital Photograph		The compressed, resized JFIF photograph is no longer stored in the barcode

Identification Code	4	*HEADER DATA:* Indicates population category. See following tables
Ver	1	*HEADER DATA:* The version of the PDF417 bar code for the form type shown by the Identification Code
PDF417 Size	2	*HEADER DATA:* Length of PDF417 bar code following this field. This value plus 7 should equal the total length of the bar code read
Checksum	1	*HEADER DATA:* Checksum of the bytes in the bar code
RSize	1	*HEADER DATA:* Indicates the size of the RAPIDS text data contained in thePDF417 bar code
DEERS Dependent Suffix/ DEERS Dependent Suffix Code	2	e.g. DDS: “20” / DDSC: “00”
Height	2	See DODI 1000.13
Weight	3	See DODI 1000.13
Hair Color	2	See DODI 1000.13
Eye Color	2	See DODI 1000.13
Direct Care Flag	1	S/N/R – See Appendix C of DoD ID Card Bar Codes & Software Development Kit
CHAMPUS Flag	1	M/N/F - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Commissary Flag	1	Y/N/X - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
MWR Flag	1	Y/N/X - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Exchange Flag	1	U/N/L/X - See Appendix C of DoD ID Card Bar Codes & Software Development Kit
CHAMPUS Effective Date	9	YYYYMMDD Format
CHAMPUS Expiration Date	9	YYYYMMDD Format.
Form Number	6	See Table 5 of DoD ID Card Bar Codes & Software Development Kit
Sponsor Person Designator Identifier (PDI)	9	9 digit PDI in the range of 000-00-0000 to 999-99-9999 was stored as a 6 digit base 32 number.
Security Code	4	Machine generated
Service/component Code	1	See Appendix C of DoD ID Card Bar Codes & Software Development Kit
Sponsor Status	6	See DODI 1000.13

Appendix A outlines the new PDF417 barcode formats for all DoD ID Card two-dimensional barcodes. To support application testing and development, samples of the changes are provided in Appendix B.

4. Timeframe

Changes to the PDF417 barcode were planned to be implemented coinciding with the Code 39 barcode changes. However, it was discovered that the elimination of the SSN from the family member PDF417 would leave the barcode without a unique identifier linking them back to the sponsor from whom the benefits and entitlements are derived. This issue has since been resolved by adding the sponsor's EDIPI number to the family member's PDF417 barcode. This change was crucial to maintain customer/patient support within the Defense Health Agency (DHA). Starting in October 2017, DoD ID cards will begin to be produced with the new bar code formats.

5. Questions and Technical References

Further information about these changes, including technical details about the barcode implementation is available upon request. Please email inquiry to the CAC Support Mailbox at cacsupport@mail.mil.

Appendix A: PDF417 Barcode Formats

The format and brief description of the contents are shown in Tables 1, 2, and 3. The DoD EDIPI, DoD Benefits Number, Unique Card Identifier and three date fields are compressed. Instructions for decompressing these fields and factors for converting the date fields to common date formats are provided in the DoD ID Card Barcodes & Software Development Kit, version 7.6.0.

(U/FOUO) Table 1 CAC PDF417 Barcode

Field	Size	Description
Barcode Version Code	1	"M"
DoD EDI Person Identifier	7	Stored as 7 digit base 32 number Range 1000000000 - 9999999999
DoD Benefits Number	8	Stored as 8 digit base 32 number Range 00000000000 - 99999999999
Person First Name	20	
Person Middle Name	1	
Person Surname	26	
Date of Birth	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Country of Citizenship	3	2 Character ISO Country Code. Will update to 3-char in the future
Personnel Category Code	1	
Branch Code	1	
Personnel Entitlement Condition Type	2	
Rank	6	
Pay Plan Code	2	
Pay Grade Code	2	
Card Instance Identifier	1	
Card Issue Date	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Card Expiration Date	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Unique Card Identifier	6	Stored as 6 digit base 32 number Range 999000000 - 999999999
Total	99	

(U/FOUO) Table 2 Sponsor ID PDF417 Barcode

Field	Size	Description
Bar Code Version Code	1	"4"
Sponsor Flag	1	"1" Sponsor
DoD EDI Person Identifier	7	Stored as 7 digit base 32 number Range 1000000000 - 9999999999
DoD Benefits Number	8	Stored as 8 digit base 32 number Range 00000000000 - 99999999999
Person First Name	20	
Person Middle Name	1	
Person Surname	26	
Date of Birth	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Country of Citizenship	3	2 Character ISO Country Code. Will update to 3-char in the future
Personnel Category Code	1	
Branch Code	1	
Personnel Entitlement Condition Type	2	
Rank	6	
Pay Plan Code	2	
Pay Grade Code	2	
Card Instance Identifier	1	
Card Issue Date	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Card Expiration Date	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Unique Card Identifier	6	Stored as 6 digit base 32 number Range 999000000 - 999999999
Total	100	

(U/FOUO) Table 3 Dependent ID PDF417 Barcode

Field	Size	Description
Bar Code Version Code	1	"4"
Sponsor Flag	1	"0" Dependent
DoD EDI Person Identifier	7	Stored as 7 digit base 32 number Range 1000000000 - 9999999999
DoD Benefits Number	8	Stored as 8 digit base 32 number Range 00000000000 - 99999999999
Person First Name	20	
Person Middle Name	1	
Person Surname	26	
Relationship	2	Person Association Type Code
Date of Birth	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Country of Citizenship	3	2 Character ISO Country Code. Will update to 3-char in the future
Sponsors DoD EDI Person Identifier	7	Stored as 7 digit base 32 number Range 1000000000 - 9999999999
Sponsor's First Name	20	
Sponsor's Middle Name	1	
Sponsor's Surname	26	
Sponsor's Deceased Flag	1	Y or N to indicate if sponsor is deceased or not, originally obtained from Status field
Sponsor's Personnel Category Code	1	
Sponsor's Branch Code	1	
Sponsor's Personnel Entitlement Condition Type	2	
Sponsor's Rank	6	
Sponsor's Pay Plan Code	2	
Sponsor's Pay Grade Code	2	
Card Instance Identifier	1	
Card Issue Date	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Card Expiration Date	4	Date converted to Julian; Subtract 2,086,303; then store as base 32
Unique Card Identifier	6	Stored as 6 digit base 32 number Range 999000000 - 999999999
Total	157	

Appendix B: Sample Barcodes

CAC PDF417

(U/FOUO) "M0TVJ0H60760I37KJOHN RDOE AUH9US AF00AMN ME02PB9NGBAPNTON1U0"



SPONSOR PDF417

(U/FOUO) "410TVJ0H70GG19S7KMICKEY MMOUSE B12FUS RA00CPT MO03LBABJBBDQTON1U1"



DEPENDENT PDF417

(U/FOUO) "400TVJ0H80GG19S7LMINNIE MMOUSE AAAT0FUS 0TVJ0H7MICKEY MMOUSE NRA00CPT MO03HBAHPBBJVTPLHGP"

